

The Cost of Capital

- Capital is inflow.
- Cost is outflow
- * Projections (cash flow estimations) - Forecast
- * No source of financing → A/P

Next
Page
↑

Value of an asset → PV of cashflows the asset generate in current period.

WACC (weighted average cost of capital)

- * weights are always determines on market rate.

$$WACC = w_d r_d (1-T) + w_p r_p + w_c r_c$$

$\downarrow$ cost of debt $\downarrow$ cost of preferred stock
weight of debt weight of preferred stock

weight of common stock
cost of common stock

- bank loan not includes $(1-T)$.

Value of share outstanding $\times$ PAR value

- Value of debt (market value)
- Retained earnings (opportunity ~~value~~ cost)

- * Dividend is your individual payment.

Cost of common stock $\rightarrow$ CAPM (required rate of return)
 $\rightarrow$ DCF (expected rate)
 $\rightarrow$ Bond yield + RP

CAPM

$$r_s = r_f + (r_m - r_f) \beta_L$$

r_f $\downarrow$ risk free rate
 r_m $\uparrow$ market rate
 $(r_m - r_f)$ $\rightarrow$ risk premium
 β_L $\rightarrow$ Beta (benchmark) - no unit

DCF

$$r_s = \frac{D_1}{P_0} + g$$

D_1 $\rightarrow$ expected and future dividend
 P_0 $\rightarrow$ current and last year dividend
 g $\rightarrow$ growth rate
 TERM = Time period

$$D_1 = D_0 (1 + g)$$

D_0 $\downarrow$ current and last year dividend

* dividend is suppose to be fixed in case of PS.

* Yield to maturity $\rightarrow$ holds to interest rate term to maturity.

* Periodic payment (compounding)

Payment $\rightarrow$ coupon rate
 Projections $\rightarrow$ Controllable
 Non-controllable

- Interest rate
- cost of debt
- Tax

Hedge $\rightarrow$ avoid risk in future.

Lec:02

Introduction

* Shareholder wealth maximization $\rightarrow$ difference between trade

* Reason for having a hierarchy in company $\rightarrow$ owner of the company
 $\rightarrow$ Trigger down the flow of command.

$\rightarrow$ CFO - looks into finance.
 Board of directors $\rightarrow$ CEO
 $\rightarrow$ COO

measure movement with respect to movement.
 management

- Beta (share price volatility)

PECRA $\rightarrow$ credit rating agency

$$\beta = 1$$

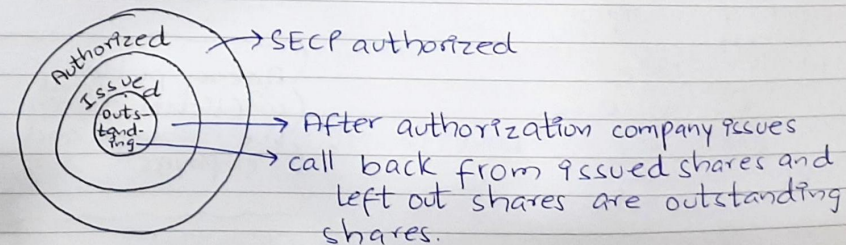
$\beta < 1$ underperform

$\beta > 1$ outperform

KPMG $\rightarrow$ Audit Grant

* Role of FM is to improve the value of shareholder wealth price.

* Financial statements portray your positions in business.



no ownership
 coupon payments
 ↑ no debt security — no liability paid.

Stocks and Bonds

↓
 ownership

PS bonds → not decision makers

↓
 cumulative not cumulative

Common bonds → decision makers

Liquidation

- ① Debt
- ② PS
- ③ Common stock

Forms → Sole Proprietorship

- ownership ①
- unlimited liability
- no legal laws involved

↓
 Corporations

- limited liability
- transfer of ownership is easy
- Law is involved

↓ Partnership

- more than 1 owner
- Agreement — Sleeping
- unlimited liability

Agency problem of interest conflicts between the manager and employees.

Finance → Investment

↓ FM ↓ capital market

Intrinsic (True) value = Market Value.

Lec: 03

Capital Budgeting

Sources of financing — Debt
 Bonds
 Equity

Debts → Bonds
 Equity → shares

* In case of equity, you can stop dividend for 1 - 2 years.

Security Valuation

PV of Inflow > Investment cost (WACC)

Steps of capital budgeting

- 1) Estimate of cashflow
- 2) Assess the riskiness of CF.
- 3) Determine Cost of capital (WACC)
- 4) Determine NPV, IRR

Projects

↓
 Independent Project

↓
 Mutually exclusive

• CF₀ is always negative because it is investment and the time period is zero.

PV of Inflow > Investment Cost (WACC)

NPV
IRR
MIRR
Payback
Discount Payback

Comparing growth
and
competitiveness.

Long term expenditure
less time period = 5 years
5-10 years

* Capital budgeting is also called
long term planning and strategic
business plan.
(decisions made by top management
CEO, CFO, board of directors.)

NPV (net present value)

$$NPV = \sum_{t=0}^n \frac{CF_t}{(1+i)^t} \quad \boxed{i = WACC}$$

* In case of mutually exclusive we will
select NPV that contributes more.
NPV which is greater.
Bench mark → Positive

If cashflow is same. } discounting.

$$CF \left[1 - \frac{i}{(1+i)^n} \right] - Cmt$$

∵ CFs are return.

Cross over rate - In case of mutually exclusive

To resolve this conflict

$$NPV_M > NPV_N$$

$$IRR_M < IRR_N$$

Reasons

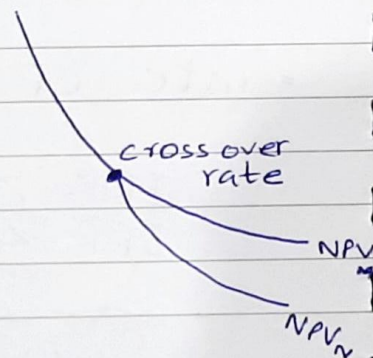
- Timings of CFs.
- Size or magnitude of the project.

Discounting

(backward)

Compounding

(forward)



Final verdict is NPV.

NPV is most accurate in opposite to IRR, MIRR, Payback etc

NPV which is greater it is selected.

Cross over rate

- Timeline of both project should be same.
(Lifespan, time, tenure)

Years	M	N	Change
0	-100	-100	0
1	10	20	10
2	40	80	40
3	30	60	30

unequal - no annuity (PMT)
 each year → annuity
 deposit → FV

Payment → Annuity payments.

Future

PMT

Present

FV

PV (rate, nper, pmt (FV) [type])

Ordinary (end of period) = 0
 (start of period) = 1

'-' before formula.

nper, rate → specify what to
 outflow and
 what to inflow.

* No FV in case of annuity.

→ fn (all formulas)

NPV

EAR
 effective rate = EAR.
 → effect

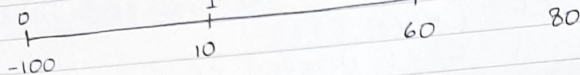
PV = Perpetuity = $\frac{\text{Annuity}}{\text{rate}}$

Time value of money
 not included
 compounding
 effect.

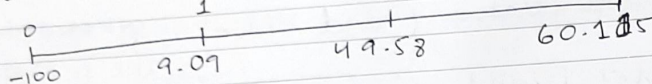
nominal rate
 stated rate

Date:
 From future to
 present.

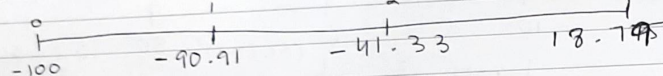
Discounted Pay back (DPBP)



PV of CF



Cumulative CF



$$2 + \frac{41.33}{60.105} = 2.68 \text{ years.}$$

* Discounted pay back is more
 realistic. (opposite to pay back period)

→ In terms
 of discounting more value of DPBP.

overstates

Lec: 07

10-09-24

Cash flow estimation

- * Usually monthly or yearly cash flow is estimated.
- * Daily basis is very difficult.
- * Incremental cashflow $\rightarrow$ cashflow that (Replacement project) affects your existing business
- * Change between Old & new cashflow.
 $\Delta \text{old} - \text{new}$

Expansion Cashflows $\rightarrow$ Cashflow that are entirely for new business.

no payback
 $\uparrow$

$\rightarrow$ Sunk Cost (cost that cannot be recovered. Once it incurs you cannot do anything about it.)

$\downarrow$
In case of expansion or incremental sunk cost become part of initial cashoutflow.

$\rightarrow$ Opportunity cost

Sales - revenue - T.C

$\uparrow$

Cashflow estimation

net operating working capital

$$\textcircled{1} \Delta \text{NOWC} = \text{CA} - \text{CL}$$

$\textcircled{2}$ Cash outflow
(Initial Cashoutlay CF_0)
 $t=0$

$\textcircled{3}$ Depreciation expense
(depreciation schedule)

$\textcircled{4}$ Cashflow estimation

$\textcircled{5}$ Terminal value - value of asset at the end of its useful life.
(Salvage value)
(residual value)

$\textcircled{6}$ Screening Tools

NPV, IRR, MIRR,
PBP & DPBP

$\textcircled{7}$ Decision (based on screening tool)

1) WACC

2) cashflow estimation

* we don't deduct interest expense and dividend PMT in cashflow estimation as we discount at WACC (which already deducts int. exp.)

* we won't include sunk cost improvement cost will be the sunk cost in the next year & it won't be included.

* Depreciation is non cash expense. Final stage (add back depreciation.)

*CAPEX → capital asset pricing expansion model. Date: _____

$$\begin{aligned} \textcircled{1} \Delta \text{NOWC} &= \text{CA} - \text{CL} \\ &= 25000 - 5000 \\ &= \$20000 \end{aligned}$$

$$\textcircled{2} \text{CF}_0 \rightarrow \text{Initial cash outlay}$$

Equipment cost	(200,000)
Installation cost	(40,000)
CAPEX	<u>(240,000)</u>
ΔNOWC	<u>(20,000)</u>
CF_0	<u>(260,000)</u>

$$\textcircled{3} \text{Depreciation schedule}$$

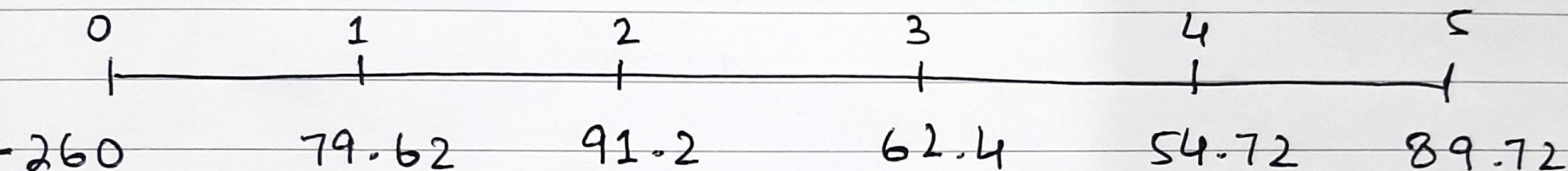
Year	Rate	Equipment Price	Depreciation amount
1	33%	240	79.2
2	45%	240	108
3	15%	240	36
4	7%	240	16.8
			<u>240</u>

$\textcircled{4}$ Operating CF's

	0	1	2	3	4
Revenue (100 @ 2)		200	200	200	200
Operating exp		(120)	(120)	(120)	(120)
dep cost		(79.2)	(108)	(36)	(16.8)
EBIT		0.8	(28)	44	63.28
Tax amount (40%)		(0.32)	(-11.2)	(17.6)	(25.28)
EBIT (1-T)		0.48	(16.8)	26.4	37.92
+ depreciation exp		79.2	108	36	16.8
		<u>79.68</u>	<u>91.2</u>	<u>62.4</u>	<u>54.72</u>
		CF_1	CF_2	CF_3	CF_4

Date: _____

Terminal cashflow	CF_4
Salvage (\$)	25
Tax rate (40-%)	(10)
After tax salvage value	<u>15</u>
+ $\Delta NOWC$	<u>20</u>
	35
	<u>54.72</u>
CF_4	<u><u>89.72</u></u>



Years

Mac A

Mac B

0

- 50,000

- 50,000

1

17,500

34,000

2

"

27,500

3

"

"

4

"

"

$$NPV_A = \$5472.68$$

$$NPV_B = \$3636.36$$

0

1

2

3

4

\$-50,000

\$34,000

\$27,500

~~\$(50,000)~~

\$34,000

\$27,500

\$-50,000

\$34,000

\$22,500

\$34,000

\$27,500

$$NPV_B = \$6,641$$

↳ on extended basis

Wednesday
26-09-24

Sensitivity Analysis

Controllable & non-controllable variables

Risk $\rightarrow$ return

Date: _____

- ① Standalone risk $\rightarrow$ entitled to an individual project
(easy) $\uparrow$ $\uparrow$ To reduce risk (diversification)

It does not check diversification.

- ② Corporate risk $\rightarrow$ due to diversification corporate risk fluctuates.
(difficult) * within an industry
* within a product line $\leftarrow$

For example

- * Limestone is the chief product of cement
- • Hedging (settle future transactions on today's price / value)

Measurements

Return $\rightarrow$ (NPV)

Risk $\rightarrow$ (standard deviation)

Coefficient of variable / correlation $\rightarrow$
relation between two variables.

- ③ Market risk

Parameters

$\uparrow$ platform, bucket of 100 well-capitalized company.
Benchmark — KSE 100

Index is the main stable
Institute.

Sensitivity Analysis

Date: _____

Total share outstanding $\times$ $\rightarrow$
Free Float $\checkmark$

Free Float = Total share outstanding $\left(\begin{array}{l} \text{No. of shares} \\ \text{outstanding} \end{array} \right)$

Beta [Performance measure of your stock with relation to the market.]

< 1 under perform
 $= 1$ parallel to market
 > 1 over perform
 $\downarrow$ out perform
realistic not possible.

Cost $\uparrow$ NPV $\downarrow$
IR $\uparrow$ NI $\downarrow$ IE $\uparrow$

- * NPV (dependent)
- * Every variable is independent

Scenario analysis

- divide NPV on scenario.
- coefficient of variation (determine)

- Quantify
- diversify

Dep. Tax saving = $\Delta \text{Dep. exp New \& Old.}$

Replacement Analysis

Step: 1 Initial cashflow CF_0 \$

Purchase

Price of new machine (8000)

Sale of old machine 2500

Tax @ Salvage Value

$(2500 - 2100) = 400 \times 40\%$ (160)

$\Delta \text{NOWC} = CA - CL = 2000 - 500$ (1500)

CF_0 \$ (7160)

Step: 2 Revenue \$

Increase in Sales 1000

Decrease in operational exp 1500

\$ 2500

Revenue After tax (2500

$\times (1 - 0.4)$)

\$ 1500

Step: 3 Depreciation

Years	1	2	3	4	5	6
Dep. of new machine	8000(20%) 1600	8000(32%) 2560	8000(19%) 1520	8000(12%) 960	8000(11%) 880	8000(6%) 480
Dep. of old machine	(350)	(350)	(350)	(350)	(350)	(350)
$\Delta \text{ in Dep exp}$	1250	2210	1170	610	530	130
Dep. Tax saving @ 40%	500	884	468	244	212	52

Date: _____

Step 4

$$\begin{aligned} \text{Opportunity cost} &= \text{New Machine Selling Price} - \text{Old machine Salvage value} \\ &= \$ (800 - 500) = \$ 300 \end{aligned}$$

Step 5

Operating CF_s estimation

Years	0	1	2	3	4	5	6
Investment							
Revenue	1500	1500	1500	1500	1500	1500	1500
Dep. Tax saving	500	500	884	468	244	212	52
ΔNWC							1500
S.V of new machine							800
Tax on SV of new machine							(320)
Opportunity cost							(300)
	0	1	2	3	4	5	6
	-7160	2000	2384	1968	1744	1712	3232

$$\text{NPV} = \$ 921.35$$

* New Machine will be acquired for MIRR is greater than WACC and NPV is positive.

Date: _____

Cost	Probability	NPV (\$)
	0.05	(70)
	0.20	(25)
	0.50	12
	0.20	20
	0.05	30

$$\text{Return} = P_1(\text{NPV}_1) + P_2(\text{NPV}_2)$$

$$= (3.5) +$$

$$\text{Risk} = \sigma^2 = P_1(x_1 - \bar{x}) + P_2(x_2 - \bar{x})$$

$$\text{C.V.} = \sigma / \text{EP} \rightarrow \sigma = \sqrt{P_1(x_1 - \bar{x})^2 + P_2(x_2 - \bar{x})^2}$$

0	-105	•
1	100	132
2	100	132
3	100	132
4	100	132
5	100	132
6		132
7		132
8		132
9		132
10		132

Capital Adequacy → end of year
ital so
that asset
is not
sold.

Tuesday

Date:

Equity Valuation

→ determine the intrinsic value of stock

Purpose

Market Value = Intrinsic Value

True Value

depends upon financial health of company.

- * Technical Analysis → Charts & graphs
- * Fundamental Analysis → studied in previous finance course.

Stock Price → Div/CF
→ Capital gain

Models

1) Dividend Discount model (DDM) → g is constant

$$\frac{D_1}{(1+r)^1} + \frac{D_2}{(1+r)^2} + \frac{D_3}{(1+r)^3} + \dots + \alpha$$

D_0 = current dividend / last year

D_1 = Future dividend / expected

2) constant growth model $r_s > g$

$$P_0 = \frac{D_1}{r_s - g}$$

cost > growth
realistically it
can be zero but
not negative.

3) Zero growth model → it holds true in the case of preferred stock. (dividend is fixed.)
 $P_0 = \frac{D_0}{r_s}$

4) Non-constant growth model
→ multiple growth rates are accounted in this model.

Problem → $\frac{r_s}{g}$ growth $\downarrow$ $\sum$ Time period $\uparrow$

CF

Free Cash flow to equity

Free cash flow to the firm

↓ FCFE
discount
@ cost of
equity

FCFF + debt repayment + debt issuance

FCFF

NI + dep - ΔNOWC - CAPEX

$$FCFE = NI - Dep - \Delta NOWC - \text{Capital expenditure} - \text{Debt repayment} + \text{Debt insurance}$$

$$\text{Capital expenditure} = \frac{\text{New Property Plant equipment}}{\text{Old PPE}} + Dep$$

Dividend — negative
Capital — positive (shown)

$$D_1 = D_0(1+g) \quad \frac{D_1}{1+g} = D_0$$

$$D_0 = \frac{78.6}{(1+0.12)} = 70.17$$

Date: _____

FCFE = \$78.6 Million
2019

Discount Model

$g_1 = 12\%$ For 2 years

$$2020 \quad 12\% \quad 78.6(1.12) = \$88.032$$

$$2021 \quad 12\% \quad 88.032(1.12) = \$98.595$$

$$\begin{aligned} r_s &= r_f + (r_m - r_f)\beta \\ &= 0.07 + (0.04)1.3 \\ &= 0.142 \\ &= 14.2\% \\ \frac{88.032}{(1+0.142)^1} &= 77.085 \end{aligned}$$

$$\frac{98.595}{(1+0.142)^2} = 75.600$$

$$\$152.685$$

Declining Growth model

$$2022 \quad 10\% \quad 98.595(1.1) = \$108.45$$

$$2023 \quad 9\% \quad 108.45(1.09) = \$118.21$$

$$2024 \quad 8\% \quad 118.21(1.08) = \$127.66$$

$$\frac{108.45}{(1+0.142)^3} = 72.816$$

$$\frac{118.21}{(1+0.142)^4} = 69.500$$

$$\frac{127.66}{(1+0.142)^5} = 65.724$$

$$\$208.045$$

Constant growth model

$$2025 \quad 7\% \quad \frac{127.66(1.07)}{0.142 - 0.07} = 1897.16$$

$$\frac{1897.16}{(1+0.142)^5} = 976.72$$

$$\text{Total} = 152.685 + 208.045 + 976.72 = 1337.45$$

$$\text{Zero} = P_0 = \frac{D_0}{r_s} = \frac{70.17}{0.142} = \$494.154$$

of which is always than BV of WACC.

To achieve optimal capital structure market should be effective.
Date: _____

adjust accordingly.

Most important factor → Market stability

Target Debt Ratio — minimum level of debt.

- WACC is always at higher level.
 $MV \text{ WACC} > BV \text{ WACC}$

$$WACC = w_d r_d (1 - T) + w_c r_c$$

risk ↑ return ↑ Cost of equity ↑

resulting to increase
MV of WACC

*debt → interest bearing
liability

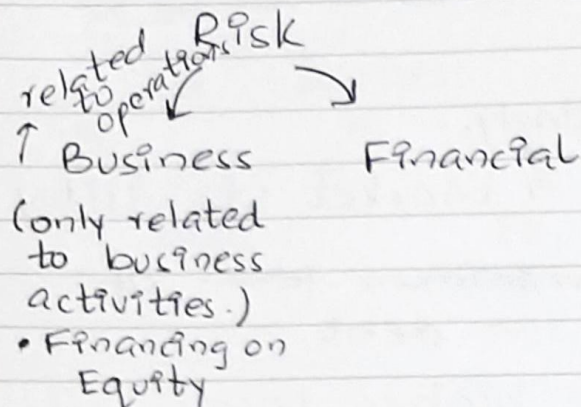
Bond is affected by

Share Price ↓
due to market
volatility.

IR ↑ IE ↑
NI ↓

Incase of CB
↓
callable
bonds.
↓
reissue
at low
rate → $r_d \downarrow$ WACC ↓

- Debt holders are the risk factors.



$$ROIC = \frac{EBIT(1-T)}{\text{Investor supplied capital}} \quad \left[\begin{array}{c} \text{Total} \\ \text{invested} \\ \text{capital} \end{array} \right]$$

$\rightarrow$ no debt on companies

$$\boxed{ROE = ROIC}$$

debt = 0

Levered = Debt inclusion

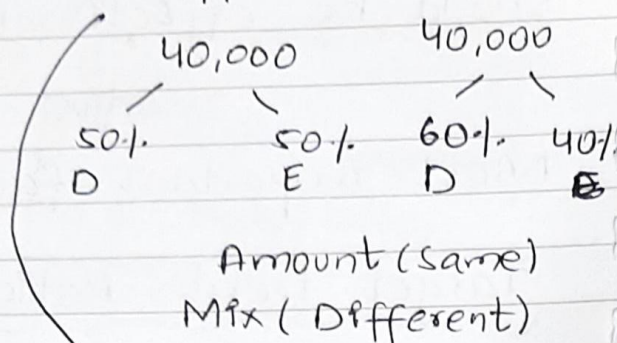
Unlevered = Debt exclusion

Demand $\uparrow$ VC $\uparrow$
and vice versa

Factors affecting Business risk

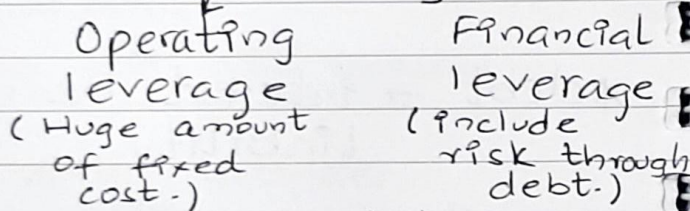
- Operations Competition
- Demand volatility
- Uncertainty about output price
- Uncertainty about output cost
 $\rightarrow$ To avoid this $\rightarrow$ Hedging (pay in advance)
- Obsolete
- Regulatory risk - Banking
- BASEL I, II, III

If the businesses have the same EBIT and investor supplied capital is same $\rightarrow$ ROIC would remain the same because

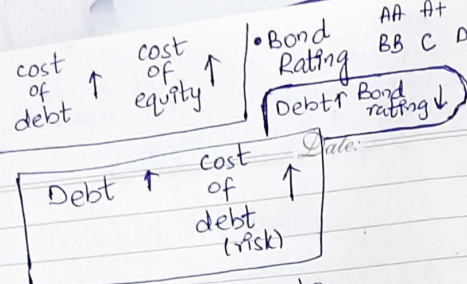


You are taking earnings before debt.

Leverage



Debt $\uparrow$
Financial leverage $\uparrow$



Recapitalization

Authorized share → legal authorization of limited shares issued.

Issued share → shares issued from authorized shares

outstanding share →

Repurchase shares to finance debt. ↑ no. of shares outstanding

Repurchase → Market capitalization ↓

→ To buy the new debt (issues new bond)

$$\text{No. of share repurchase} = \frac{\text{New Debt Issued}}{\text{Price Per share}}$$

Amount Borrowed	Debt to Capital ratio	Debt to equity ratio	Bond rating	rd
\$0	0	0	-	-

\$250,000 0.125 0.143 AA 8%.

\$500,000 0.250 0.333 A 9%.

EBIT = 8,400,000

Tax rate = 40%.

D = 250,000

Price Per share = \$25

No. of shares = 80,000

EPS = ?

TIE = ?

$$\text{EPS} = \frac{(\text{EBIT} - r_d D)(1-T)}{\text{No. of shares outstanding}} = \frac{(8400000 - 0(0))(1-0.4)}{80000(25)} = 2.52 = \$3$$

$$\text{Issues repurchase} = \frac{250,000}{25} = \$10,000$$

$$\text{Price Per share} \xrightarrow{25} \text{EPS} = \frac{8400000 - 0.08(250,000)(1-0.4)}{80000 - 10000(25)}$$

$$\text{EPS} = \$3.25$$

$$\text{TIE} = 20 \times$$

$$r_d \times D = \text{Int. Exp}$$

$$D = \$500,000$$

$$\text{EPS} = 3.55$$

$$\text{TIE} = 8.8 \times$$

$$\frac{\text{EBIT}}{\text{Int. exp}}$$

Question

Existing capital structure

$$D = 20\% \quad E = 80\%$$

Future capital structure

$$D = 40\% \quad E = 60\%$$

$$r_f = 5\% \quad , \quad r_d = 8\% \quad , \quad r_m - r_f = 6\% \quad , \quad r_s = 12.5\%$$

$$\text{Tax rate} = 40\%$$

$$\text{WACC} = ?$$

Solution First WACC

$$\text{WACC} = w_d r_d (1-T) + w_s r_s$$

$$\begin{aligned} &= (0.2)(0.08)(1-0.4) + 0.8(0.125) \\ &= 0.1096 \\ &= 10.96\% \end{aligned}$$

Second levered Beta

$$r_s = r_f + (r_m - r_f) \beta_L$$

$$\beta_L = \frac{r_s - r_f}{(r_m - r_f)} = \frac{0.125 - 0.05}{0.06} = 1.25$$

Third Unlevered Beta

$$\beta_L = \beta_U [1 + (1-T) D/E]$$

$$1.25 = \beta_U \left[1 + (1-0.4) \frac{0.2}{0.8} \right]$$

$$\beta_U = \frac{1.25}{1.15} = 1.087$$

Now New β_L

$$\beta_L = \beta_U [1 + (1-T) D/E]$$

$$= 1.087 \left[1 + (1-0.4) \frac{0.4}{0.6} \right] = 1.52$$

$$\text{New } r_s = r_f + (r_m - r_f) \beta_L = 0.05 + (0.06) 1.52 = 14.12\%$$

$$\text{New WACC} = w_d r_d (1-T) + w_s r_s$$

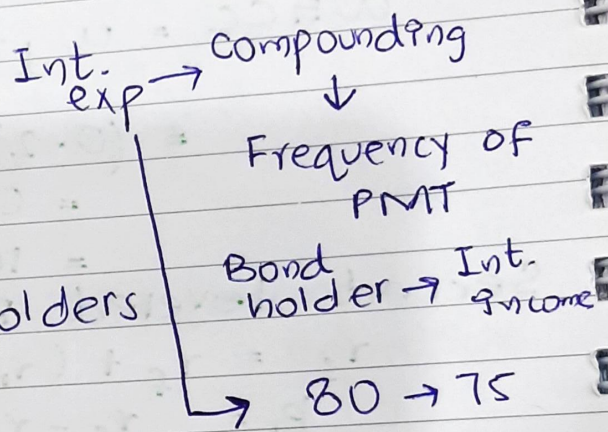
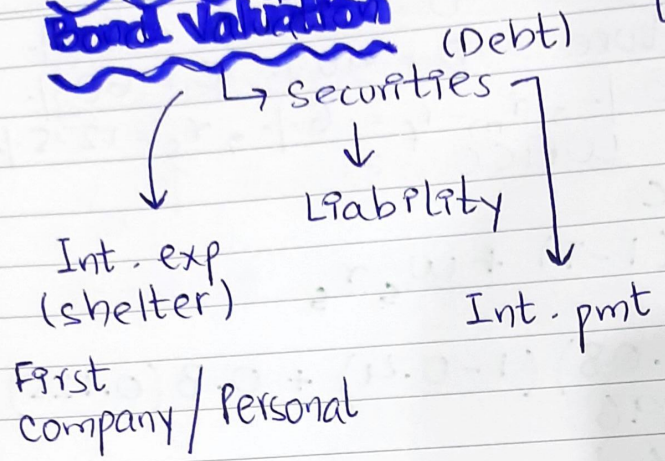
$$\begin{aligned} &= (0.4)(0.08)(1-0.4) + (0.6)(0.1412) \\ &= 10.392\% \end{aligned}$$

→ Opt Recapitalization

Date: _____

Bond Valuation

[Issued by the company to issue the financing for the firm.]



- Tax benefit for bond holders



* gov projects are on debt financing.

Debt to equity ratio ↑

Bond rating ↓
Default risk ↑

r_d ↑ - in order to compensate the bond holder

Bond rating
(Impact on value of bond)
PECRA Question

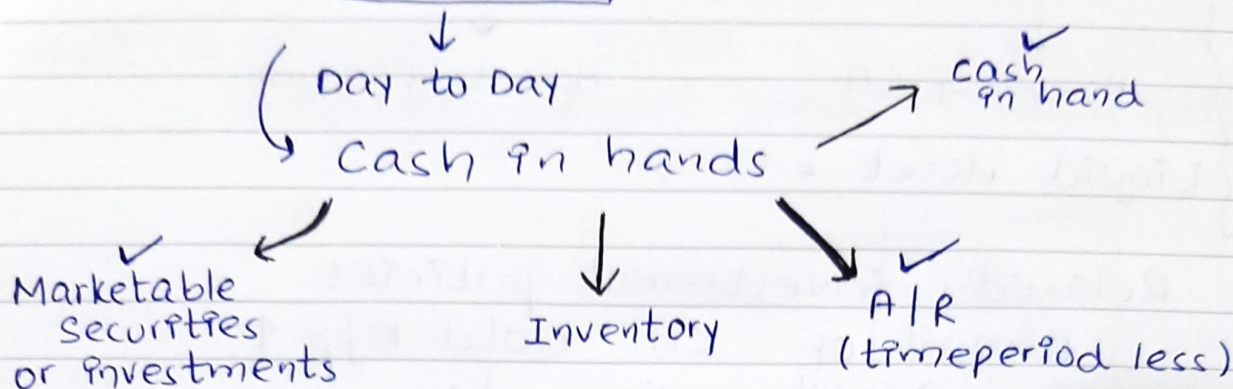
* with time period volatility increases.

(Effects by)

- Time period
- Inflation
- Bond rating

[Finals] Chap : 16 Working Capital Manag.

Date: 06-11-24



A/P = Rs 25000 → Liability
once you pay this it becomes expense.
↓
After 25 days

Double Cycle

Time value of money → Money at hand is better than money in future.

* Marketable Securities → Bonds, stocks etc.

(1) Net working capital = CA - CL

(2) Net operating working capital = (CA - CL) - Note Payable

↓
operations / operational activities

↓
Why?
Financing activities (interest rate)

* Current Asset investment policies

Amount of CA
How to finance
Liquid Asset =

① Relaxed investment policies
Amount of CA hold A/R ↑
Credit policy relax

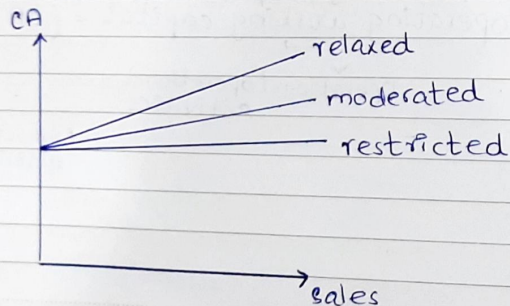
③ Profitability ↑
① Receivables ↑
② Turn over time ↓

* NI is cash from sale

② Restricted Investment policies
Best examples (Banks)

Deposits (For banks Liability)
- Reserves

③ Moderate Investment Policies



Date:

Duopond Analysis of ROE

$$ROE = \text{Profit Margin} \times \text{Total Asset Turn over} \times \text{Equity Multiplier}$$

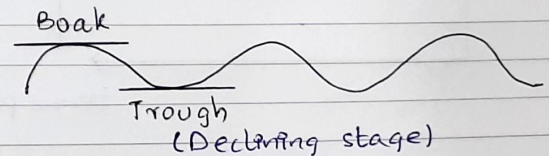
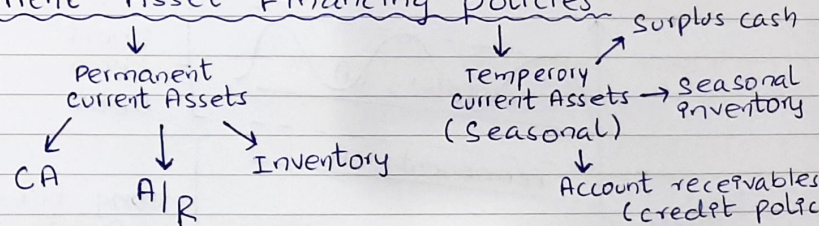
$$= \frac{NI}{\text{Sales}} \times \frac{\text{Sales}}{TA} \times \boxed{\frac{TA}{\text{equity}}} \rightarrow \text{Financing}$$

$$ROE = \frac{NI}{\text{equity}} \times \frac{\text{Sales}}{TA}$$

(productivity) (old)
→ what proportion of my income is my income.

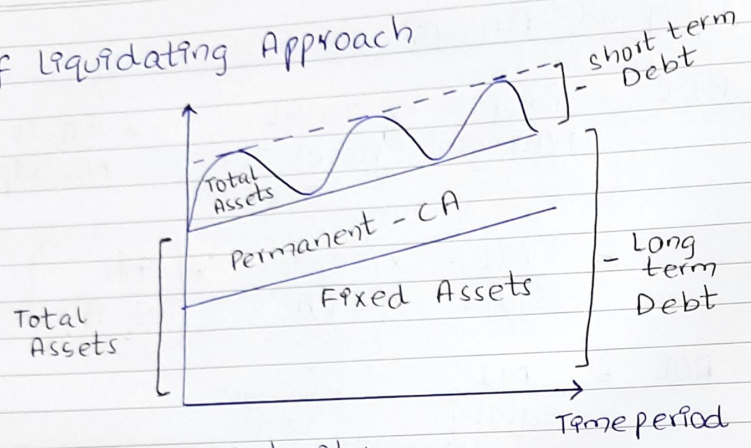
* ROE is the end result of your financing.

Current Asset Financing policies



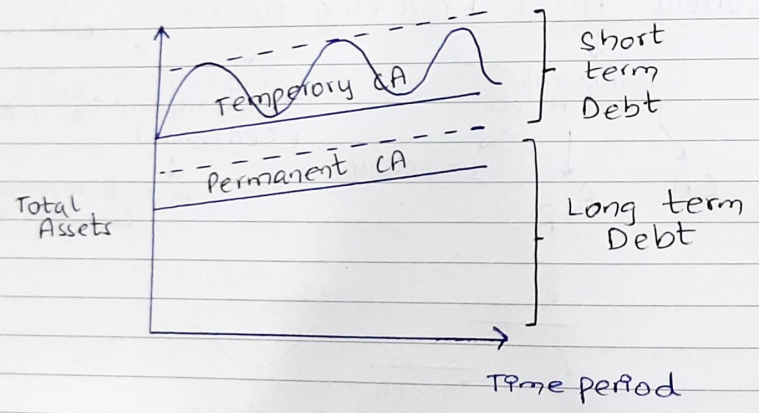
Date: _____

① Self liquidating Approach



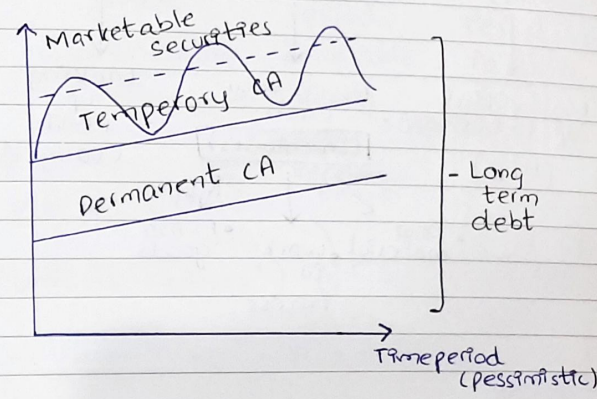
* Time period of Maturity of financing and maturity of assets are same.

② Restricted Approach



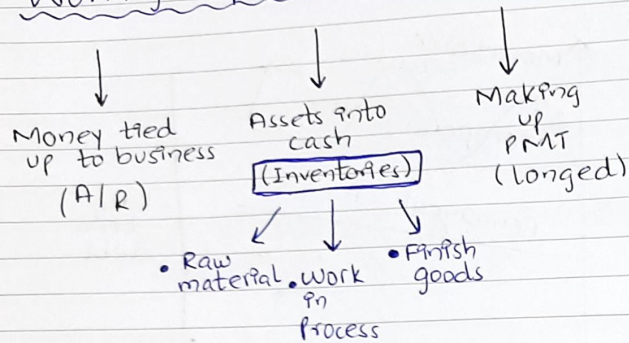
Date: _____

③ Conservative Approach



Date: 12-11-24

Working Capital Management → Payable



* Credit terms → Apply to payables and collection (receivables)

e.g. $\frac{2}{10} n 30$
 ↓
 credit period

Discount 2% if pays within 10 days.

Material worth = \$10,000

Date:

$$\text{Cash Conversion Cycle} = CCC = \frac{\text{Inventory}}{\text{Conversion Period}} + \frac{\text{Receivable}}{\text{collection Period}} - \frac{\text{Payable}}{\text{Defferal Period}}$$

$$= \frac{\text{Inventory}}{\text{COGS} / 365} + \frac{\text{Receivable}}{\text{Sales} / 365} - \frac{\text{Payables}}{\text{COGS} / 365}$$

$$\text{Inventory turnover} = \frac{\text{COGS}}{\text{Inventory}}$$

→ (unit is 'times')

Receivable turnover = $\frac{\text{Sales}}{\text{Inventory}}$

Ask Ma'am

List Price

↓

Price = \$100

↗

= \$98

Face Value

② Discount charge [discount Period]

1 - 10

↓
In this period we are saving

Free cost of trade credit

Other way of financing
 - cash freed up

11 - 30

↓
Time is cost of trade credit

For calculation of comparison.

Nominal Cost of Trade Credit (Forgo interest rate)

$$= \frac{\text{Discount Rate}}{100\% - \text{Discount rate}} \times \frac{365}{\text{No. of trade Period (Total)} - \text{Discount Period}}$$

* In case above 30 days either 31 or even 35 days. The days will be 30.
→ you won't deduct it from 10.
→ as no discount is availed.

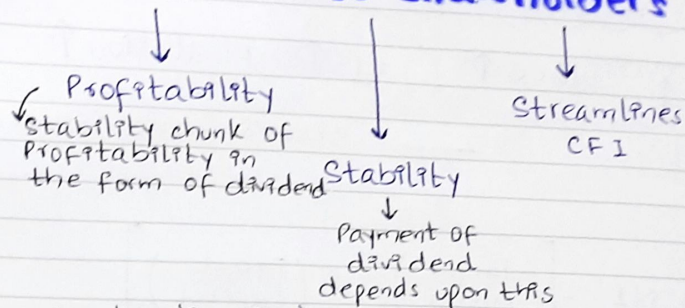
$$\text{Effective Cost of trade credit (Company)} = \left[1 + \frac{\text{Discount rate}}{100 - DR} \right]^{\frac{365}{n - DR}} - 1$$

$$\text{Effective Cost of trade credit (Bank)} = \left[1 + \frac{i_{nom}}{m} \right]^m - 1$$

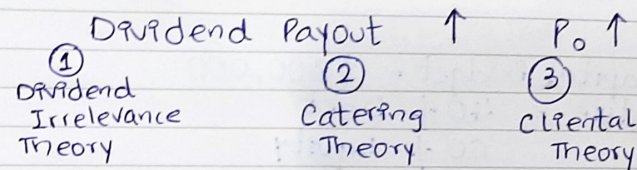
Date:

Chap: 15 Distribution to shareholders

Date: 21-11-24



* Declaration Date



Originate from MM Theory (no tax - unrealistic)

* Market is efficient.

* Investor is indifferent to the dividend.

* Investor looking for stable income prefer dividend.

* dividend Policies should be modified.

Dividend residual Model

Cost of capital $\uparrow$ WACC $\uparrow$ r_d $\uparrow$ Floatation $\uparrow$
_____ shares more issues.

$$\text{Dividend} = \text{NI} - \left[\left(\frac{\text{Target equity ratio}}{\text{ratio}} \right) \times (\text{Target capital}) \right]$$

Question $\rightarrow$ Capital Budget = \$800,000
Capital Structure = 40% Debt
60% Equity

$$\text{NI} = \$600,000$$

How much of the forecasted NI is paid at dividend = ?

$$\text{DIV } P_0 = \frac{\text{DIV}}{\text{NI}}$$

$$\begin{aligned} \text{DIV} &= 600,000 - [60\% \times 800,000] \\ &= 120,000 \\ \text{DIV } P_0 &= \frac{120,000}{600,000} = 20\% \end{aligned}$$

Date: _____

$$\text{If } \text{NI} = 400,000 - [60\% \times 800,000]$$

$$\begin{aligned} \text{DIV} &= 400,000 - [60\% \times 800,000] \\ &= -80,000 \end{aligned}$$

$$\text{DIV } P_0 = 0 \quad (\text{due to negative})$$

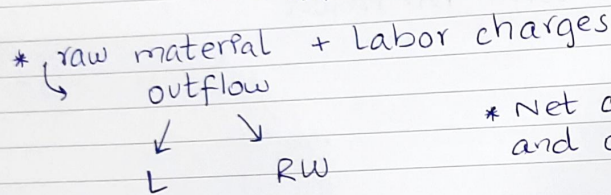
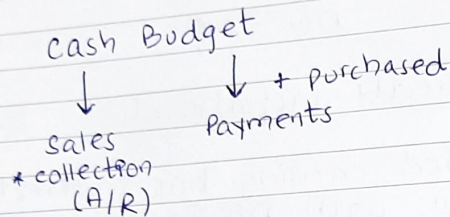
$$\text{NI} \uparrow \quad \text{DIV } P_0 \uparrow$$

Directly Proportional

$\rightarrow$ Retained earning has indirect relation with DIV Payout.

Chap: 16 Cash Budget

- * Made at the end of every month.
- * Deals with inflows
- * Depends on credit terms (different for inflows and outflows)



* Net cash inflows and outflows.

- * Must pay special attention to each credit terms.

Date: _____

Cash Budget

Date: _____

May June July Aug Sep Oct Nov Dec

Sales

COLLECTIONS

"

"

"

Total collections

May June July Aug Sep Oct Nov Dec

PAYMENTS

Purchases

PMT

NET CASH GAIN OR LOSS

July Aug Sep Oct Nov Dec

Collections (inflow)

PMT

G & A

Leases

Miscellaneous expense

Income tax

PMT

Repair (R & P)

+ Net cash

Cash in hand

Cumulative Cash

- Target

+

* negative value means incomplete target.

Question

2018 and 2019
May 2018
June
July
August
September
October
November
December
January 2019

\$
180,000
180,000
360,000
540,000
720,000
360,000
360,000
90,000
180,000

collected within the month of sale = 10%
collected the month following the sale = 75%
collected the second month following the sale = 15%
Labor + Raw material

\$
May 2018 40,000
Jun 40,000
July 126,000
August 882,000
September 306,000
October 234,000
November 162,000
December 90,000

General and administrative salaries = \$27,000
Lease payments = \$9,000
Miscellaneous expenses = \$2700
Progress Payment = \$180,000 in Oct
Cash on hand = \$132,000 on 1st July
Depreciation charges = \$36,000
Income tax payments = \$63,000 are due in Sep & Dec.
Minimum cash balance = \$40,000

* 1st month value → 2nd Month Value
• we add from July as all credit terms received from that month.
• we are making 6 month cash Budget.

Answer

Cash Budget

	May	June	July	Aug	Sep	Oct	Nov	Dec
Sales	180	180	360	540	720	360	360	90
Collections								
Collection within the month of sale 10%	18	18	36	54	72	36	36	9
Collection the month following the sale 75%		135	135	270	405	540	270	270
Collection in second month 15%			27	27	54	81	108	54
Total Collections		198	351	531	657	414	333	
Payments								
Purchases	90	90	126	882	306	234	162	90
PMT		90	90	126	882	306	234	162
Net Cash Gain or loss								
Collection (inflows)	198	351	531	657	414	333		
PMT	90	126	882	306	234	162		
G & A	27	27	27	27	27	27		
Lease	9	9	9	9	9	9		
Miscellaneous expense	27	27	27	27	27	27		
Income Tax PMT	-	-	-	63	-	63		
Repair (R&P)	-	-	-	180	-	-		
Net Cash	69.3	186.3	(452.7)	132.3	141.3	69.3		
Cash in hand	132	201.3	387.6	(65.1)	67.2	208.5		
Cummulative Cash	201.3	387.6	(65.1)	67.2	208.5	277.5		
Target	90	90	90	90	90	90		
	111.3	297.6	(155.1)	(22.8)	118.5	187.5		

* Cash inflows should be increase or cash outflows.
* lacking target cash in these 2 months

Chapter 8 15

Dividend & share repurchase

Date: _____

Setting dividend policy

- ① Forecasting capital over horizon period
- ② Set the target capital structure
- ③ ~~Divide~~ Decide annual equity (Retained earnings)
- ④ Residual payout of Div
- ⑤ Growth rate

Stock split:

(No. of shares increase,

Price decrease

but total worth will remain same.)

↓
5 for 1

3 for 2

→ When stock price is high, stock split is done so individuals can invest easily in the company.

No. of investors ↑

• 1 share for \$25 / each

after stock split 1 share for \$5 / each
so 5 people will buy 5 shares
for \$25.